```
# OSL PRACTICAL 5
# Shell Scripting

cd ~/OSL_Practicals
mkdir -p Practical5
cd Practical5

# 1. Multiplication table using FOR, WHILE and UNTIL

cat > multiplication.sh << 'EOF'
#!/bin/bash

echo "Enter a number:"
read num

echo "Using FOR loop:"
for ((i=1; i<=10; i++))
do
    echo "$num x $i = $((num*i))"
done

echo "Using WHILE loop:"
i=1
while [ $i -le 10 ]
do
    echo "$num x $i = $((num*i))"
    ((i++))
done

echo "Using UNTIL loop:"
i=1
until [ $i -gt 10 ]
do
    echo "$num x $i = $((num*i))"
    ((i++))
done
EOF

# 2. Positive, negative or zero

cat > number_check.sh << 'EOF'
#!/bin/bash

echo "Enter a number:"
read num

if [ "$num" -gt 0 ]
then
    echo "The number is POSITIVE."
elif [ "$num" -lt 0 ]
then
    echo "The number is NEGATIVE."
else
    echo "The number is ZERO."
fi
EOF

# 3. File existence and information
```

```
cat > file_info.sh << 'EOF'
#!/bin/bash

echo "Enter file name:"
read filename

if [ -e "$filename" ]
then
    echo "File exists."
    echo "File information:"
    ls -l "$filename"

    echo "File size:"
    du -h "$filename"

    echo "Detailed information:"
    stat "$filename"
else
    echo "File does not exist."
fi
EOF

# Give execute permission
chmod +x multiplication.sh number_check.sh file_info.sh

# Execute scripts one by one
./multiplication.sh
./number_check.sh
./file_info.sh

# Display all files
ls -l
```